# Notes on Formalizing PLN Evidence Algebra and the Markov de Finetti Theorem

(Lean/Mettapedia technical summary; archived failed-attempt record)

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**Abstract**

This note is archived as a record of a formalization direction that was not completed to a final accepted proof package.

These notes summarize two intertwined formalization threads inside the `Mettapedia` Lean project:

1. A clean algebraic and categorical packaging of *probabilistic logic network* (PLN) "evidence" and deduction rules via an *evidence quantale*, with a bridge to enriched-category style composition.
2. An attempted Lean formalization of a *Markov de Finetti / Diaconis–Freedman* theorem on finite state spaces: Markov-exchangeable prefix measures satisfying a recurrence hypothesis are mixtures of Markov chains.

The purpose of the note is didactic: it records the core mathematical ideas, the engineering decomposition used in Lean, and the blocker that caused archival before completion of the hard-direction proof.

**Status note.** Implementation-status and count claims in this draft are snapshot-scoped; verify against current repository trackers and build outputs before citing as current state. Lean-proven claims are only those tied to explicit Lean file/theorem references; broader mathematical or historical claims are literature-backed context and not asserted as Lean-proven unless explicitly stated.


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# 1 Motivation and the "Solomonoff ↔ probability theory" theme

Solomonoff induction is usually presented as a universal Bayesian mixture over computable hypotheses (Solomonoff 1964). In practice, one often *restricts* to structured hypothesis classes (e.g. i.i.d. models, finite-state Markov models, hidden Markov models) both for tractability and for interpretability. The conceptual bridge in this project is:

> *Exchangeability (or partial exchangeability) assumptions identify when a rich family of distributions can be represented as a mixture over a simpler parametric family.*

For i.i.d. sequences, de Finetti gives exchangeability ⇒ mixture of i.i.d. laws (de Finetti 1931/1937). For Markov sequences, Diaconis–Freedman show a corresponding representation for Markov-exchangeability (partial exchangeability of trajectories), with an additional structural recurrence hypothesis (Diaconis & Freedman 1980).

Once such structure theorems are formalized, they can be used to justify *structured universal mixtures*: Solomonoff-like prediction can be instantiated by mixing over the parametric space of Markov models, with hyperpriors that retain dominance-style guarantees in the restricted model class.

# 2 Prefix measures and (partial) exchangeability

## 2.1 Prefix measures on a finite alphabet

Let $A$ be a finite alphabet (in the Lean development, $A = \mathrm{Fin}(k)$). A *prefix measure* is a function

$$\mu : A^* \to [0, 1]$$

satisfying the cylinder-consistency equations (informally: $\mu(xs) = \sum_a \mu(xs \cdot [a])$ and $\mu([]) = 1$). Such a $\mu$ can be viewed as the finite-dimensional marginals of a probability measure on infinite trajectories, when an extension exists.

### 2.2 Markov exchangeability

For Markov chains, a finite trajectory $x_0, x_1, \ldots, x_n$ has natural sufficient statistics:

- the starting state $x_0$,
- the transition-count matrix $N(i,j) = \#\{t < n : x_t = i, x_{t+1} = j\}$,
- (optionally) the last state $x_n$.

A prefix measure is *Markov-exchangeable* when $\mu(xs)$ depends only on these statistics, not on the detailed order in which transitions occur. This is a form of partial exchangeability: permutations that preserve the transition counts preserve probability.

## 3 Markov parameter space and word probabilities

### 3.1 Parameter space

For a finite state space $\mathrm{Fin}(k)$, a Markov model can be parameterized by:

- an initial distribution $\pi_0$ on $\mathrm{Fin}(k)$;
- a row-stochastic transition kernel $P(i, \cdot)$ for each $i \in \mathrm{Fin}(k)$.

The Lean development packages this as a compact topological space

$$\Theta = \mathrm{MarkovParam}(k),$$

essentially a finite product of simplices. Compactness is important: it enables the use of Stone–Weierstrass density and the Riesz–Markov–Kakutani representation theorem.

### 3.2 Word probability as a continuous function

Given $\theta = (\pi_0, P) \in \Theta$ and a finite word $xs = [x_0, \ldots, x_n]$, define

$$\mathrm{wordProb}_\theta(xs) = \pi_0(x_0) \cdot \prod_{t=0}^{n-1} P(x_t, x_{t+1}).$$

In Lean this is implemented as a measurable/continuous kernel

$$\theta \mapsto \mathrm{wordProb}\ \theta\ xs,$$

with a real-valued continuous coercion `wordProbReal`. This function lies in the coordinate-generated subalgebra of $C(\Theta, \mathbb{R})$.

## 4 Evidence partitions as a Markov analogue of Bernstein bases

A key engineering move is to avoid reasoning about individual words directly, and instead regroup words by their Markov sufficient statistics.

Fix a horizon $n$. There are finitely many trajectories of length $n+1$ over $\mathrm{Fin}(k)$. Each trajectory has an evidence summary $e$ (start, counts, last).

### 4.1 Two parallel families indexed by evidence

For each $n$ and evidence class $e$:

- $w_\mu(n, e)$ is the total mass assigned by $\mu$ to trajectories with evidence $e$.
- $W(n, e) : \Theta \to [0, 1]$ is the total probability under $\theta$ of all words in that evidence class:

$$W(n, e)(\theta) = \sum_{\text{traj } xs:\ \text{evidence}(xs)=e} \text{wordProb}_\theta(xs).$$

Crucially, for each fixed $n$, the finite family $\{W(n, e)\}_e$ forms a *partition of unity* on $\Theta$:

$$\forall \theta, \quad \sum_e W(n, e)(\theta) = 1,$$

and $\{w_\mu(n, e)\}_e$ is a probability vector.

This is the Markov counterpart of the role Bernstein basis polynomials play in one-dimensional Hausdorff moment proofs: they turn global approximation/representation questions into finite-dimensional simplex constraints at each $n$.

## 5 Recurrence as an identifiability hypothesis

### 5.1 Definition (prefix-measure recurrence)

The Diaconis–Freedman recurrence condition can be phrased on an extension $P$ to infinite trajectories:

$$P\{X_n = X_0 \text{ i.o.}\} = 1.$$

In the Lean development, recurrence for a prefix measure $\mu$ is defined as: there exists an extension measure $P$ to $(\mathbb{N} \to \text{Fin}(k))$ whose cylinder sets agree with $\mu$ and for which the recurrence event holds almost surely.

### 5.2 Why recurrence is necessary

Markov exchangeability alone does *not* imply mixture-of-Markov-chains. A fully formal counterexample is constructed: a deterministic chain that leaves its start state once and never returns. It is Markov-exchangeable as a prefix measure, but violates recurrence, and therefore cannot be represented as a mixture over Markov parameters.

Interpretation: recurrence is an *identifiability anchor*: it guarantees the trajectory keeps resampling transitions from a reference state, allowing long-run transition statistics to stabilize. Without it, transient drift can preserve partial exchangeability on finite prefixes while breaking any global mixture interpretation.

## 6 The Markov de Finetti theorem and the Lean proof architecture

### 6.1 Target statement (informal)

Let $\mu$ be a Markov-exchangeable prefix measure on $\text{Fin}(k)$ satisfying recurrence. Then there exists a probability measure $\Pi$ on $\Theta = \text{MarkovParam}(k)$ such that for every word $xs$,

$$\mu(xs) \ = \ \int_\Theta \text{wordProb}_\theta(xs)\, d\Pi(\theta).$$

Equivalently: $\mu$ is a mixture of Markov chains.

### 6.2 Functional-analytic wrapper

The formalization follows a compactness/functional-analysis template:

1. Define the map $\Pi \mapsto \left(\int W(n,e)\, d\Pi\right)_{(n,e)\in u}$ for a finite set $u$ of constraints.
2. Show this map is continuous, and that the space of probability measures on $\Theta$ is compact.
3. Define the *moment polytope* as the image of this compact set; it is compact and hence closed.
4. Reduce the hard direction to a *finite satisfiability* statement: for each finite $u$, the constraint vector defined from $\mu$ lies in the moment polytope.

Intuitively, the hard work is proving that the finite-dimensional constraints implied by Markov exchangeability + recurrence are consistent with some mixing measure.

## 7 The remaining "Diaconis–Freedman core" approximation lemma

### 7.1 What remains

At the time of writing, the Lean development has reduced the entire hard direction to a single approximation lemma (referred to as `good_state_bound` in the code). Conceptually, it compares:

- a *without-replacement* distribution arising from uniform sampling over trajectories in a fiber determined by a Markov state summary, and
- a *with-replacement* product distribution induced by the empirical Markov parameter of that state summary.

The desired inequality has the form

$$|W(\text{empiricalParam}(s)) - \text{prefixCoeff}(s)| \;\le\; \frac{C}{M},$$

where $M$ is the number of returns to the anchor state (or a lower bound thereof), and $C$ is a constant depending only on $(k,n)$ (and the chosen evidence granularity), not on $s$.

### 7.2 Excursion decomposition strategy

The main combinatorial device is an *excursion decomposition*: a trajectory is cut into segments between consecutive returns to the start state. The project formalizes:

- return positions and counts,
- the resulting list of excursions,
- the induced "uniform-on-fiber" measure as a *sampling without replacement* model on excursion lists.

The bound is then obtained by:

1. proving a one-step (per-excursion-prefix) deviation bound between without-replacement and with-replacement probabilities (already formalized in the excursion model files),

2. lifting to a length-$m$ prefix bound using a generic product perturbation inequality:

$$\left|\prod_{i=1}^{m} p_i - \prod_{i=1}^{m} q_i\right| \leq \sum_{i=1}^{m} |p_i - q_i|,$$

3. summing over all excursion-prefix events compatible with the evidence partition.

The only remaining engineering step is a clean decomposition lemma expressing both `prefixCoeff` and `W(empiricalParam s)` as finite sums over excursion-prefix events, so the already-proven excursion bounds can be applied termwise.

# 8 PLN evidence as an algebraic/categorical object

Separately from the Markov de Finetti work, the project develops a robust algebraic view of "evidence" used in probabilistic logic networks.

## 8.1 Evidence counts and projections

Evidence is represented as a pair of nonnegative counts $(e^+, e^-)$. Two common projections are:

- strength, typically $e^+/(e^+ + e^-)$ (when total evidence is nonzero),
- confidence, typically a monotone function of total evidence $e^+ + e^-$.

These projections connect to the more traditional PLN truth value representation (strength, confidence), but the evidence-pair view is often algebraically cleaner.

## 8.2 Quantale structure and residuation

A central insight is that evidence pairs form a commutative monoid under a tensor-like operation (roughly: coordinatewise multiplication in a suitable semiring/complete-lattice setting), and this can be upgraded to a *quantale* with a right adjoint (residuation). This structure supports:

- a compositional view of implication chaining,
- a "direct path" vs "indirect path" decomposition of deduction (via complements and residuation),
- a bridge to enriched-category composition laws.

The Lean development contains a theorem explicitly connecting the PLN deduction lower bound to an enriched composition law.

# 9 Measure-theory curriculum (minimal toolkit that repeatedly mattered)

For an assistant (human or agent) joining the formalization effort, the following results tend to be the practical "spine":

1. **Cylinder sets and Kolmogorov-style consistency:** how prefix measures relate to measures on infinite products.

2. **Tonelli/Fubini for nonnegative integrals:** commuting $\int$ with finite sums.
3. **Continuity of integration:** $\Pi \mapsto \int f \, d\Pi$ is continuous for continuous bounded $f$ on compact spaces.
4. **Compactness of probability measures on compact spaces:** enabling finite intersection arguments and closed-image (moment polytope) arguments.
5. **Stone–Weierstrass:** density of coordinate-generated subalgebras inside $C(\Theta)$.
6. **Riesz–Markov–Kakutani:** turning positive linear functionals on $C(\Theta)$ into measures (when that route is used explicitly).

Notably, the current Markov hard-direction path leans more on finite-dimensional compactness + continuity + closed-image reasoning than on a full Daniell–Stone extension, reserving RMK for the final measure extraction step.

## 10 Roadmap for closing the last lemma

To finish the hard direction in Lean (and thereby close the Solomonoff–Markov exchangeability bridge in this thread), the remaining work is sharply focused:

1. Introduce a small refactor to avoid an import cycle: move the statement of the remaining bound into a file that can import both the excursion model and the approximation wrapper.
2. Prove two decomposition lemmas:
   - `prefixCoeff` is a finite sum over excursion-prefix coefficients.
   - `W(empiricalParam s)` is the matching finite sum over with-replacement excursion probabilities.
3. Apply the already-proven excursion prefix bound termwise, then aggregate using the generic product-difference inequality.

## Acknowledgments / provenance

This note is extracted from a collaborative formalization thread and the corresponding Lean source files in the `Mettapedia` project. It is intended as a technical memory aid for future contributors.